

Gaussian Quadrature in Lean and Rocq

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Abstract

We formalize Gaussian quadrature in Lean, starting from Appel and Bindel’s preliminary Rocq development and comparing the proofs needed to connect real analysis, floating-point arithmetic, and C execution. Using `mathlib`, `LeanPDE`, and `FloatLib`, we construct weighted Gaussian rules of every positive order on compact intervals, prove their remainder and convergence, and certify binary64 results for all ten rules stored in the example program. Checking the example exposes two incorrect error bounds: 0.00223 for two nodes and 0.02 for one node. We prove the replacements 0.00356 and 0.159, correct the remainder’s inconsistent node indexing, model the weight actually stored in C, and give counterexamples to three admitted helper lemmas. For the C proof, we replace the external cosine with a verified polynomial. For each stored rule, every permitted evaluation order of an initialized call then terminates with the certified result in our Lean adaptation of CompCert’s C semantics. We prove that translation to Clight preserves these calls, and that the C return values agree with the result annotations of ten assembly applications verified directly in Lean. Independent Rocq proofs use CompCert’s compiler theorem. The Lean result concerns these particular programs; general correctness of source translation and compilation, and agreement between the Lean and Rocq definitions of execution, remain open.

1 Introduction

Appel and Bindel [2] study the verification of a C program that approximates

$$\mathcal{I}(f) = \int_{-1}^1 f(x) dx, \quad f(x) = \frac{1}{2}(1-x)\cos x, \quad (1)$$

using Gaussian quadrature. The integrator occupies seven lines of C. Yet proving its accuracy requires real analysis, floating-point error bounds, memory and function-call reasoning, and a theorem about compilation. This makes it a useful example for comparing Lean and Rocq: the program is small enough to inspect, while its correctness depends on all of these proofs.

Their paper is titled *Formalization of Gaussian Quadrature and Application Verification (Preliminary Draft)*. Its admitted lemmas mark unfinished work. We review that preliminary version and the accompanying public source, distinguishing false statements from missing proofs and unassembled program modules. The source already contains work on correcting the derivative-bound hypotheses; we discuss its status in Section 2.5. Our results are in a separate development and do not complete the authors’ Rocq files.

Their Rocq development brings together MathComp and MathComp-Analysis [13, 1] for real analysis, Flocq [4] for floating-point arithmetic, LAProof [10] for rounding error in sums, Interval [14] for concrete inequalities, and VST [5] for C verification. CompCert’s compiler theorem [12] provides the next connection, from verified source behavior to formal assembly. Our development, LeanQuadrature, uses `mathlib` [17], `LeanPDE` [11], and `FloatLib` [8] for the mathematical and

numerical proofs. For execution, we adapt CLean [6] and prove the required properties directly from the language semantics.

The example also lets us check its accuracy claims independently of the program proof. The integral is $\sin 1$, and the exact two-point rule returns $\cos(1/\sqrt{3})$. Their difference is about 0.0035591571, which already exceeds the manuscript’s bound of 0.00223. The one-point bound is also false. We prove corrected bounds, identify inconsistent indexing in the admitted remainder theorem, and use the actual C weight where the original floating-point model differs. Section 2 gives the calculations and counterexamples. It also distinguishes three false admitted helper lemmas from unfinished program obligations, such as the missing callback body proof and module assembly. An admitted lemma is an assumption: checking a proof that uses it does not establish that assumption.

Our mathematical results apply beyond this integrand. For every positive order, we construct a Gaussian rule on a compact interval with a positive continuous weight. We prove exactness, positivity of the weights, the three-term recurrence, the integrated remainder, and convergence for continuous integrands. The original development supplies concrete root certificates through four nodes and leaves the general remainder admitted. Our floating-point results cover all ten rules stored in its C source, with error bounds for the actual constants and rounded operations. Appendix A maps all 24 numbered paragraphs of their mathematical presentation to the results available in LeanPDE and identifies the declarations used by LeanQuadrature.

At the C level, we replace the external cosine with a polynomial whose C body and numerical use are both verified. We prove that initialized calls to the integrator terminate with the certified result in every evaluation order permitted by our adapted semantics. We then prove that the six selected library functions preserve their call results when translated to Clight, CompCert’s simpler intermediate language for C. The application programs report the same binary64 values and exit zero; we prove this behavior through formal assembly in Lean. We also prove those application results independently in Rocq, where they compose with CompCert’s compiler theorem.

The C library and assembly applications expose their results differently. A library call returns a value without producing an event. The application records that value in a semantic annotation, then exits. We prove that these observations agree for all ten orders, including the original two-point wrapper. This connects the results of the particular programs. It does not prove that compiling the original C text produces the imported assembly trees. General source-translation and compiler correctness, and agreement between the adapted Lean and Rocq execution rules, remain open.

1.1 The five levels of verification

The five levels are the real integral, the exact quadrature rule, its floating-point model, the C implementation, and the compiled program. Four proofs connect adjacent levels. Write $\mathcal{Q}_n(f)$ for the exact quadrature sum and F for the binary64 result of the rounded functional model. The first two connections establish

$$|\mathcal{I}(f) - \mathcal{Q}_n(f)| \leq E_{\text{quad}}, \quad |\mathcal{Q}_n(f) - \text{real}(F)| \leq E_{\text{fp}}. \quad (2)$$

The first inequality bounds quadrature error over the reals. The second accounts for stored constants, callback error, and rounding. The triangle inequality then gives

$$|\mathcal{I}(f) - \text{real}(F)| \leq E_{\text{quad}} + E_{\text{fp}}.$$

The third proof shows that the C function returns F , with valid memory accesses and calls. The fourth preserves that source behavior through compilation. These last two proofs introduce no new numerical error; their task is to preserve the computation already analyzed.

The cosine implementation affects what can be concluded at the C level. The original program calls an external cosine, so its verification needs a contract for that implementation. Our replacement polynomial has a verified C body and certified numerical results at the inputs used by the example. We also prove initialization of the stored tables. The resulting library theorems therefore need neither an assumed cosine contract nor an assumed table predicate.

Table 1 compares this work with Appel and Bindel’s development. The independent Rocq application proofs that we use with CompCert are described separately in Section 6; they are part of LeanQuadrature, not changes to the authors’ Rocq proofs.

Reading the labels. ✓ **Proved** means the stated theorem is checked under its hypotheses. ● **Partial** and ● **Conditional** mean an application still has obligations to discharge. ○ **Open** means the connection is not proved. ✕ **Refuted** means a counterexample disproves the claim.

LeanPDE supplies the weighted Gaussian theory, explicit Legendre rules through four nodes, and general root and weight certificate proofs. These results build on mathlib and LeanPDE’s quadrature interfaces, midpoint rule, and two-point rule. Rational interval arithmetic, Taylor bounds, and integration tactics provide concrete enclosures. Its sharp two-point error bound, proved with a Peano kernel, and its general Hermite remainder give two proofs of the fourth-derivative bound for that rule. LeanQuadrature contains the stored tables, their particular root brackets, numerical accuracy certificates, and program proofs. FloatLib supplies the binary64 operations and rounding theorems.

2 Corrections to the example and its proofs

Both the integral and the ideal quadrature sum have closed forms, so we can check the claimed errors directly. A quartic tests the remainder formula, and decoding the table literals tests whether C and the rounded model store the same values. Table 2 records these four findings. The rest of this section examines the admitted helpers and program specifications on which the application proof depends. These are corrections to a preliminary development. The absence of a proof is a coverage question; a counterexample establishes the stronger claim that a statement must change before it can be proved. Page references and source claims concern the examined preliminary manuscript and accompanying source [2, 3]; the repository records their exact versions and maps each finding to its declarations.

2.1 The two-point error bound

Page 5 states the bound $223/100000 = 0.00223$ for the exact-real two-point rule. Symmetry in (1) gives

$$|\mathcal{I}(f) - \mathcal{Q}_2(f)| = |\sin 1 - \cos(1/\sqrt{3})| \approx 0.0035591571.$$

The error is larger than the proposed bound. The Lean proof works with rational enclosures for sine and cosine, so the displayed decimal is only an illustration. An upper enclosure for $\sin 1$ and a lower enclosure for $\cos(1/\sqrt{3})$ also give the corrected bound 0.00356. Listing 1 shows the final proof. The first rewrite replaces the integral and the sum by their closed forms, and the two remaining goals are linear inequalities between previously proved enclosures.

The C corollary on pages 1 and 11 uses the slightly larger bound 0.00224 for the reported result 0.83791182769499317. Its analytic premise therefore needs correction before the program argument can establish an accuracy bound. The counterexample above concerns the ideal rule. To prove an error bound for a C execution, we must also account for its cosine implementation and rounded operations, as in Sections 4 and 5.

Level	Appel–Bindel in Rocq	LeanQuadrature in Lean
1. Real integral	MathComp, MathComp-Analysis: weighted real integral.	mathlib: interval integrals. LeanPDE: checked symbolic integration. ✓ Proved: the example integral equals $\sin 1$.
2. Numerical rule	MathComp, MathComp-Analysis. Partial: recurrence and quadrature proofs, with root certificates through four points. The general remainder is admitted; convergence is unproved.	mathlib, LeanPDE: the shared Gaussian theory. ✓ Proved: construction at every positive order on compact intervals; exactness, positive weights, recurrence, the corrected remainder, and convergence.
3. Rounded model	Flocq, LAProof, Interval. Partial: rounded fold and error analysis. The example’s accuracy argument uses admitted numerical helpers; two overflow helpers have false conclusions for permitted parameters.	FloatLib: binary64 arithmetic and rounding bounds. LeanPDE: root and weight enclosures. ✓ Proved: finite fold, error bounds, and ten stored rules; callback accuracy at the inputs used by the examples.
4. C execution	VST. ✓ Body proof: VST relates the integrator to its functional model under table and callback specifications. Partial: the wrapper uses admitted numerical lemmas; the callback body and assembly into a Verified Software Unit (VSU), including global initialization, remain obligations.	Adapted CLean, FloatLib, with our execution proofs. ✓ Proved: initialized calls using the original functions and a verified C cosine polynomial, at ten orders and in every permitted evaluation order. Behavioral preservation to Clight for all six selected functions. Open: general source-translation correctness and agreement with Rocq’s execution semantics. The external-cosine variant retains its contract.
5. Compilation	CompCert. ✓ Compiler theorem: CompCert preserves program behavior. Using it for this application requires source correctness in its C semantics.	Adapted CLean, with our assembly proofs. ✓ Particular programs: direct proofs for ten polynomial applications at formal assembly, reporting the values returned by the C library. Open: a verified compiler, or a transfer of CompCert’s theorem into Lean.

Table 1: Libraries and proof coverage at each level. The Rocq column describes Appel and Bindel’s preliminary development, including the admitted dependencies discussed in Section 2. Our C proofs use a verified polynomial in place of the external cosine. Their return values agree with the assembly applications’ result annotations by Theorem 6.1. That theorem concerns these programs’ behavior, not the correctness of a general compiler.

```

theorem corrected_two_point_bound :
  |(∫ x in (-1 : ℝ)..1, integrand x) -
    (integrand (-(1 / Real.sqrt 3)) + integrand (1 / Real.sqrt 3))| ≤
    (356 / 100000 : ℝ) := by
  rw [integral_eq, two_point_eq, abs_of_nonneg]
  · linarith [sin_one_upper, cos_node_lower]
  · linarith [sin_one_lower, cos_node_upper]

```

Listing 1: The corrected two-point bound, from `Quadrature.Examples.TwoPointCosine`. The enclosure lemmas are proved earlier in the same module.

Location	Finding	Correction checked in Lean
Page 5, two points	× Refuted : the ideal error bound 0.00223.	✓ Proved : the replacement bound 0.00356. Actual error ≈ 0.0035591571 .
Page 5, one point	× Refuted : the ideal error bound 0.02.	✓ Proved : the replacement bound 0.159. Actual error $1 - \sin 1 \approx 0.1585290152$.
Page 4, remainder	× Refuted : the formula combines n nodes with a derivative of order $2n + 2$ and the polynomial p_{n+1} .	✓ Proved : the remainder uses $2n$ derivatives and p_n , with C^{2n} regularity. Two nodes require the fourth derivative.
Pages 6–7, weight	✓ Mismatch proved : the C and model weights have different binary64 bits; the C value exceeds the stated tolerance.	✓ Proved : the C value satisfies a $5 \cdot 10^{-16}$ weight bound. Our initialized library uses these actual C bits.

Table 2: Corrections and their proof status. The first three rows concern real mathematics. The fourth repairs the numerical model to match the stored C constant and uses it in the initialized Lean library.

2.2 The one-point error bound

The one-point node is zero and its weight is two, so

$$\mathcal{Q}_1(f) = 2f(0) = 1, \quad |\mathcal{I}(f) - \mathcal{Q}_1(f)| = 1 - \sin 1 \approx 0.1585290152.$$

This exceeds the bound $2/100 = 0.02$ on page 5 by almost a factor of eight. We prove both the failure of that inequality and the replacement bound $159/1000 = 0.159$ in Lean.

2.3 Counting nodes and derivatives

The displayed `quadrature_error` statement on page 4 uses an n -node root collection together with a derivative of order $2n + 2$ and a polynomial indexed by $n + 1$. With n the number of nodes, the remainder requires order $2n$, as in Theorem 3.1. The source makes the node count explicit: its root certificate stores an n -tuple, and the quadrature sum ranges over $i < n$.

The formula $2n + 2$ is right for a rule with $n + 1$ nodes indexed from zero through n , which is Stewart’s convention [15]. The problem is that the node count and the derivative order follow different conventions in the same statement. For two nodes the required derivative is the *fourth*, and for one node it is the second.

A quartic makes the consequence concrete. For $f(x) = x^4$, two-point quadrature has error

$$\int_{-1}^1 x^4 dx - \left[(-1/\sqrt{3})^4 + (1/\sqrt{3})^4 \right] = \frac{2}{5} - \frac{2}{9} = \frac{8}{45},$$

while $f^{(6)}$ is identically zero. The displayed formula would give zero error. Lean proves the exact error $8/45$ and the vanishing sixth derivative, which together refute that formula for a two-node rule.

The manuscript marks the general remainder theorem as admitted. Its source statement also quantifies over an arbitrary $f : \mathbb{R} \rightarrow \mathbb{R}$ with no smoothness premise. Our theorem, `GaussianRule.remainder`, uses the derivative of order $2n$ and requires C^{2n} regularity on the closed interval. Section 3 explains the proof.

2.4 A weight differs between C and the model

The C listing on page 6 stores the outer three-point weight as 0.5555555555555556. The functional-model value on page 7 is 0.5555555555555556. In binary64 these are different numbers:

C literal	0x3fe1c71c71c71c76
Functional-model literal	0x3fe1c71c71c71c72

The C value decodes exactly to 2501999792983611/4503599627370496. Its distance from the ideal weight 5/9 is

$$\frac{19}{40532396646334464} \approx 4.6876083262 \cdot 10^{-16},$$

which exceeds 2^{-53} . So the C constant misses the stated uniform half-ulp tolerance even when half an ulp is measured at one. Lean proves this inequality and the difference between the two stored values. We keep the original C constant and prove the uniform weight bound $5 \cdot 10^{-16}$ for the stored tables instead.

The mismatch also affects the program proof. The generated Clight program contains the C bits, but the table predicate in the VST specifications describes the model’s array. A body theorem may assume that predicate, but a proof from program initialization must establish it for the array that is actually there, and with these literals it cannot. We model the existing C literals and enlarge the error budget. Our selected library initializes those values and proves the required table predicate. The original VST development would need a corresponding change to its model and initialization proof. VST’s Verified Software Unit construction checks the connection between global initializers and the module’s data predicate. The quadrature source has not yet assembled this VSU, so the successful body proofs do not assert that initialization satisfies their table premise [19]. The middle three-point weight has different decimal spellings in the two files, but both spellings round to the same binary64 value, so that entry is not a second mismatch.

2.5 Assumptions in the original proofs

The two example lemmas, `error_1_0_1` and `error_1_0_2`, use `quadrature_error_bound_is_bound`, which depends on a Legendre specialization of the admitted `quadrature_error`. Interval checks an inequality derived from the assumed remainder. The remainder itself remains an assumption, so a later proof ending in `Qed` establishes an implication from that assumption. This explains how the final scripts can compile while the claimed error bound is false: the unchecked premise already contains the error.

The C development has a similar dependency. Its integrator body proof is complete under its specification. The example wrapper invokes admitted lemmas for function bounds, derivative bounds, callback accuracy, and quadrature error. The verification file supplies nine body lemmas for ten internal functions and never proves `body_testfun`. The wrapper obtains the callback contract from the global specification environment with `make_func_ptr`, which establishes what may be assumed about the callback, not that its C body meets the contract.

To see exactly what the completed proofs depend on, we rebuilt the original quadrature proofs and their LAMeasure dependencies and asked Rocq for the transitive assumptions of 43 declarations. The wrapper depends on ten of the twelve admitted declarations in the quadrature files, including the false positivity and overflow helpers described below. The integrator body depends on none of the twelve. The rebuild needed two explicit product-type annotations in the high-level 2D specifications, which resolve a notation ambiguity in our Rocq environment and change no proof, admitted lemma, or C code. All nine original body proofs then compile. The committed Clight

syntax was generated for AArch64 with the Apple ABI, and our review checks it under an x86-64 CompCert configuration.

The preliminary manuscript acknowledges the missing remainder proof. Consequently its concluding description of a first-principles argument should be read as the intended completed chain. The checked state of that chain still includes the assumptions above.

Work already under way on derivative bounds. The public source includes a separate `proof/g_max_deriv.v` that reformulates `fbound` and `deriv_bound` on a fixed interval with $a < b$. It requires both the starting and perturbed arguments to lie in that interval. Its target bounds the change in the function by a derivative bound times the displacement. This is the appropriate mean-value argument for controlling perturbed quadrature nodes. The file is work in progress: `g_max_deriv` ends in `Qed` but uses the admitted `g_max_deriv_oneway`, and other admissions remain. It is separate from the integrated quadrature proofs in the examined revision [3]. This ongoing correction is relevant to the example’s admitted function and derivative bounds. Our accuracy theorem states the required Lipschitz bound explicitly, and the concrete numerical certificates establish their own bounds.

A positive function with zero integral. The admitted lemma `Rintegral_gt_0` assumes continuity, nonnegativity, and a nonzero value somewhere in $[a, b]$, and concludes that the integral is positive. The missing condition is $a < b$. For $a = b = 0$ and $h(x) = 1$, all three premises hold, but

$$\int_{[0,0]} h(x) dx = 0$$

because a singleton has Lebesgue measure zero. Our Lean counterexample checks the three premises and refutes the conclusion. We also check the example in Rocq with MathComp-Analysis, and a separate diagnostic applies the original admitted lemma to those premises and derives a contradiction. The source uses this helper in its polynomial-positivity argument, inside a section that already assumes $a < b$, so adding the missing premise is a natural repair for that use. Our `polynomialIntegral_square_pos` requires $a < b$ and proves the result using `mathlib`.

A specification that names the wrong function. The source also includes Hughes quadrature routines for a triangle. Its `hughes_weight_spec` begins with `DECLARE _gauss2d_weight` where the intended identifier is `_hughes_weight`. As a result, `quadrules_ASI` contains two specifications named `_gauss2d_weight` and none named `_hughes_weight`. The body proof still checks because VST’s `semax_body` receives a function body and an identifier–contract pair, checks the body against the contract, and discards the identifier. Associating names, bodies, and contracts is a separate obligation when the verified interface is assembled. A VSU checks that the specification used at a call is supplied by the corresponding body or by a contract that implies it [19]. That is the missing integration step here. The identifier can be corrected without changing the contract, and the existing body theorem then proves the retagged specification. This is a defect in the specification of the triangle routine, not a counterexample to its numerical result.

2.6 Two overflow lemmas need stronger hypotheses

The floating-point proof must show that every intermediate result is finite. The source introduces a condition named `parameter_limits` and leaves two helper inequalities admitted. Neither helper follows from the stated conditions for all permitted parameters, and the problem can be seen without running any floating-point code.

Specialize to binary64 and write

$$u = 2^{-53}, \quad \eta = 2^{-1075}, \quad T = 2^{1024}.$$

These are the relative-error constant, the absolute-error constant, and `fmax` in the version of L`A`Proof used by the source. The largest finite binary64 number is smaller than T . Take one node, the zero integrand, and a callback that always returns zero. Choose the valid but loose function bound $F = T/4$ and set the derivative and callback-error bounds to zero. The node zero and the weight two are represented exactly. The source’s range condition becomes

$$\left(1 + \frac{u}{1+u}\right)(2+u)\frac{T}{4} + \eta < T,$$

which holds, and its earlier product helper also holds. But the admitted `finite_integrate_model_aux` would require

$$\frac{T}{2} + \frac{T}{4}(u^2 + 3u) + \eta < \frac{T}{2(1+u)}. \quad (3)$$

The left side is greater than $T/2$ and the right side is smaller. This is a counterexample at a positive order, with valid bounds for both the mathematical function and the callback.

The other helper, `gauss_pt_wt_limit_aux`, fails at zero nodes, which the source’s type `'I_5` permits. With function bound $F = T$ and zero callback error, `parameter_limits` reduces to $0 < T$, while the helper requires $2T(1+u) + \eta < T$. Either the empty reduction needs separate treatment or the helper needs a positive-order premise.

Lean checks both counterexamples in `Quadrature.Examples.Overflow` with exact arithmetic on the transcribed inequalities. Both callbacks return zero, and their actual computations stay finite. The contradictions concern the proposed sufficient conditions for the helper lemmas. L`A`Proof’s summation theorem has its own valid precondition; the missing step is to derive it inside the quadrature development. Our finiteness proof, in Section 4, bounds the sum of absolute products together with the accumulated rounding errors.

3 Gaussian rules in Lean

The remainder theorem depends on the nodes, weights, and polynomial exactness of the rule. Appel and Bindel follow Stewart [15]: they construct the monic orthogonal polynomials by the three-term recurrence and take a `roots_of_ortho_p n` certificate containing n ordered roots, which their concrete Legendre instances supply through four points. We construct the orthogonal polynomials by orthogonal projection, prove the existence of their roots by factorization and integral positivity, and derive the recurrence afterward. This yields a Gaussian rule for every positive order. The construction uses `mathlib`’s polynomial algebra, orthogonal projection in finite-dimensional inner product spaces, and Lagrange interpolation. Its remainder and convergence proofs use Rolle’s theorem and uniform polynomial approximation. LeanPDE provides this general theory and the concrete Legendre rules. Appendix A maps the mathematical paragraphs to its definitions and theorems.

Fix $a < b$ and a continuous function $w : [a, b] \rightarrow \mathbb{R}$ with $w(x) > 0$. The intended integral is

$$\mathcal{I}_w(f) = \int_a^b f(x)w(x) dx.$$

For a positive integer n , a Gaussian rule consists of distinct interior nodes x_i and weights λ_i , with $\mathcal{Q}_n(f) = \sum_{i=1}^n \lambda_i f(x_i)$. Throughout the paper n is the *number of nodes*.

3.1 Choosing the nodes and weights

The integral defines an inner product on real polynomials, $\langle p, q \rangle = \mathcal{I}_w(pq)$. It is positive definite because a nonzero polynomial cannot vanish on a nondegenerate interval, so its square has positive integral against w . Orthogonal projection onto the polynomials of degree less than n constructs the unique monic polynomial p_n of degree n orthogonal to that space.

The root argument shows that p_n has exactly n distinct roots in (a, b) , and these are the nodes. Lagrange interpolation defines the weights. Polynomial division by p_n then proves exactness:

$$\mathcal{Q}_n(q) = \mathcal{I}_w(q) \quad \text{whenever} \quad \deg q \leq 2n - 1. \quad (4)$$

Writing $q = p_n s + r$ gives $\deg s < n$ and $\deg r < n$. Orthogonality kills the integral of $p_n s$, and evaluation at the nodes kills its quadrature sum.

If ℓ_i is the i th Lagrange basis polynomial, exactness applies to ℓ_i^2 and gives $\lambda_i = \mathcal{I}_w(\ell_i^2) > 0$. Applying exactness to the constant polynomial gives $\sum_i \lambda_i = \mathcal{I}_w(1)$. These two facts control perturbations of the integrand later without estimating each weight separately. The development also proves the three-term recurrence, including the positivity of the norm ratio, and identifies the explicitly represented low-order Legendre rules with the constructed monic family.

3.2 Integrating the remainder

Theorem 3.1 (Integrated Gaussian remainder). *Let $a < b$, let w be positive and continuous on $[a, b]$, let $n > 0$, and let $f \in C^{2n}([a, b])$. For the Gaussian rule constructed above, there exists $\xi \in [a, b]$ such that*

$$\mathcal{I}_w(f) - \mathcal{Q}_n(f) = \frac{f^{(2n)}(\xi)}{(2n)!} \int_a^b p_n(x)^2 w(x) dx. \quad (5)$$

In Lean the smoothness hypothesis is `ContDiffOn` on the closed interval, and the derivative is the iterated derivative within that interval, so we never assume a smooth extension beyond the integration domain. More precisely, the proof takes functions $F_0, \dots, F_{2n}$ continuous on $[a, b]$, with $F'_k = F_{k+1}$ on (a, b) for $k < 2n$. The `ContDiffOn` version supplies this chain using `iteratedDerivWithin`.

Taylor–Lagrange itself is already in the mathlib revision used here: `taylor_mean_remainder_lagrange` gives the Taylor remainder about one center. Our added interpolation theorem concerns n distinct nodes, matching the function and its first derivative at each node. It is this Hermite remainder, followed by an integral argument, that proves Theorem 3.1. We do not formalize arbitrary prescribed derivative multiplicities in the interpolant.

Let H be the Hermite interpolant of degree below $2n$ matching both f and f' at the nodes. For a point x that is not a node, choose $c = (f(x) - H(x))/p_n(x)^2$ and consider

$$G(y) = f(y) - H(y) - c p_n(y)^2.$$

There are double zeros at the n nodes and one more zero at x . Repeated Rolle supplies a zero of the $2n$ th derivative. Since $H^{(2n)} = 0$ and $(p_n^2)^{(2n)} = (2n)!$, this identifies c . At a node the pointwise remainder is zero directly. This pointwise theorem needs continuity only for F_k with $k < 2n$, together with the derivative chain in the interior. The integrated theorem additionally uses continuity of F_{2n} .

The pointwise result does not yet give the integrated remainder. Its witness ξ_x varies with x , and nothing says that $x \mapsto \xi_x$ is measurable, let alone continuous. Integrating an arbitrarily chosen witness would introduce an unjustified regularity assumption. We use its range instead. Put

$$r(x) = (f(x) - H(x))w(x), \quad \phi(x) = \frac{p_n(x)^2 w(x)}{(2n)!}, \quad g = f^{(2n)}.$$

For every x , $r(x)$ equals $g(y)\phi(x)$ for some $y \in [a, b]$. The function ϕ is nonnegative with positive integral, so the minimum and maximum of the continuous function g bound $\int r / \int \phi$, and the intermediate value theorem supplies a point where g equals that quotient. Exactness for H now gives (5). The argument depends only on the range of g . We state it as a separate integral theorem so that other remainder proofs can reuse it.

If $|f^{(2n)}| \leq M$ on the interval, the absolute error is at most $M\mathcal{I}_w(p_n^2)/(2n)!$. For unit weight on $[-1, 1]$, the coefficients of M at orders one through four are $1/3$, $1/135$, $1/15750$, and $1/3472875$.

3.3 Convergence

For a continuous f , choose a polynomial q uniformly close to f . Once the order is high enough to integrate q exactly, positivity and the constant total weight give

$$|\mathcal{I}_w(f) - \mathcal{Q}_n(f)| \leq 2\mathcal{I}_w(1) \|f - q\|_\infty.$$

Uniform polynomial approximation proves convergence. Stewart lists this as a consequence of positive weights, and the manuscript notes it as unproved because the applications did not need it. The result does not require differentiability and is distinct from the derivative-based rate in Theorem 3.1.

The program stores rules for orders one through ten. For each table we identify the ideal nodes and weights, prove enclosures for them, and bound the distance from the stored constants. These certificates let the general theory apply to the concrete program. Everything here is on a compact interval. Gauss–Hermite and Gauss–Laguerre rules would need integrability and approximation arguments on unbounded domains. They are outside our development and explicitly outside the implementation described in the preliminary paper.

4 Floating point in Lean

The integral bound concerns exact nodes and exact arithmetic. To apply it to a program we need a second model that fixes the stored values, the callback, and the order of rounding. The original Rocq model is a left fold of rounded products and sums, parameterized by a floating-point format and a NaN policy, and its analysis uses Flocq, LAProof’s dot-product results, and Interval. The Lean model uses the same evaluation order with FloatLib’s binary64 operations [8]. FloatLib is executable, so a statement about a particular rounded computation can often be settled by kernel evaluation, and it comes with rounding bounds for the general argument. Let $\hat{x}_i, \hat{\lambda}_i$ be stored nodes and weights, and let $\hat{f}$ be a floating-point callback. The computation is

$$s_0 = +0, \quad t_i = \text{fl}(\hat{\lambda}_i \hat{f}(\hat{x}_i)), \quad s_i = \text{fl}(s_{i-1} + t_i). \quad (6)$$

Each multiplication and addition rounds separately. Reassociating the sum or using fused operations would change the computation being verified.

A stored-rule certificate states that every node and weight is finite, the decoded nodes lie in $[-1, 1]$, and

$$|\text{real}(\hat{x}_i) - x_i| \leq \delta_x, \quad |\text{real}(\hat{\lambda}_i) - \lambda_i| \leq \delta_w.$$

For the ten stored tables, common bounds are $\delta_x = 6 \cdot 10^{-16}$ and $\delta_w = 5 \cdot 10^{-16}$. The latter covers the actual C literals, including the three-point constant from Section 2.

Theorem 4.1 (Accuracy of the rounded rule). *Let $n > 0$ and let a stored rule certify an ideal Gauss–Legendre rule. Suppose $|f| \leq B$, f is L -Lipschitz, $f \in C^{2n}([-1, 1])$, and $|f^{(2n)}| \leq M$. Assume*

$B, e, L \geq 0$ and that, on every finite binary64 input decoding into $[-1, 1]$, the callback returns a finite value with error at most e from f at that decoded input.

Write ε_R for FloatLib’s local absolute rounding bound at radius R . If

$$(2 + n\delta_w)(B + e) + 2n\varepsilon_R \leq R \leq \text{maxFinite}, \quad (7)$$

then the execution in (6) is finite, and

$$\begin{aligned} |\text{real}(s_n) - \mathcal{I}(f)| &\leq 2n\varepsilon_R + (2 + n\delta_w)(e + L\delta_x) + n\delta_w B \\ &\quad + \frac{M}{(2n)!} \int_{-1}^1 p_n(x)^2 dx. \end{aligned} \quad (8)$$

Each term in (8) has a distinct source. There are n products and n additions, each contributing at most ε_R . At a stored node, callback error and displacement of the node contribute at most $e + L\delta_x$, and multiplying by the total absolute stored weight gives the factor $2 + n\delta_w$. Changing the weights themselves contributes $n\delta_w B$. The last term is the exact-real quadrature error from Theorem 3.1.

The range condition (7) is what makes this a statement about finite arithmetic. It bounds the absolute mass of all products and reserves room for every rounding step. The proof establishes finiteness and the rounding estimate together. This supplies the finiteness argument for which the original development uses its admitted helpers. The resulting budget can then be discharged for each concrete application.

The callback hypothesis in Theorem 4.1 covers the whole input interval. For a fixed rule we can also prove accuracy directly at the finitely many inputs the computation uses. We do this for FloatLib’s executable cosine at the two stored nodes, with cosine error at most 10^{-15} , by evaluating the implementation in the kernel and comparing with a rational Taylor enclosure. The resulting rounded quadrature value has absolute integral error at most 0.00356. This covers FloatLib’s cosine at these inputs. A platform `libm` would need its own contract.

For the ten application proofs we use a callback that evaluates the degree-14 Taylor polynomial for cosine by a fixed rounded Horner scheme, so that no external call remains, and we certify each final result against a rational enclosure of $\sin 1$. Section 5 explains why this choice lets us discharge the callback contract in C.

5 Verifying C execution in Lean

The numerical proof bounds the error of a value F . We must now show that the C integrator returns that value. Listing 2 contains the complete function. The proof has to justify each table access and function call and show that the compound assignment performs the same rounded multiplication and addition as (6).

After k iterations, the loop invariant relates the C variable `s` to the model’s partial sum s_k , and it retains the table facts and callback specification needed for the next iteration. At loop exit $k = n$ and $s_n = F$, which connects the numerical theorem to the returned value. Appel and Bindel express this argument in VST’s `semax_body` judgment, and VST’s soundness theorem relates that judgment to CompCert’s semantics [18, 5]. Our Lean proofs work directly with execution rules, using the loop invariant in proofs about the permitted transitions.

CLean [6] is an experimental fork of CompCert whose `clightgen` can emit Clight syntax as Lean. It supplies the Clight semantics, a sound interpreter, and a separation logic with function contracts, a frame rule, and rules for function-pointer calls. Its `closure` theorem combines verified

```

double integrate (double (*f)(double), int n) {
  int i;
  double s = 0.0;
  for (i=0; i<n; i++)
    s += gauss_weight(i,n) * f(gauss_point(i,n));
  return s;
}

```

Listing 2: The original one-dimensional C integrator. The table accessors and callback are verified separately.

function bodies using a decreasing natural-number measure for recursive calls. It also provides code for linking generated modules and resolving declarations against definitions. These are inherited capabilities. Our quadrature proofs use its execution rules and interpreter soundness directly.

We adapted CLean in two ways. Its arithmetic used host floating-point operations. We replaced these with FloatLib’s definitions so that arithmetic certificates can be checked by kernel evaluation. External calls were axioms, and we replaced them with explicit environment parameters and determinism contracts, so that every external assumption appears in the theorem statement. These definitions support the execution proofs throughout the development. We also formalized CompCert’s typed C expression reductions and small-step machine in Lean, using the same value and memory definitions. The rules permit every operand order allowed by CompCert’s C semantics. This is a manual adaptation: a theorem relating its definitions to the original Rocq definitions remains open.

The safety conclusion is relative to these execution rules, including their treatment of unsafe expressions and invalid memory accesses. A proof in this model alone does not establish that a manual adaptation has preserved every relevant C behavior. The semantic correspondence remains an obligation alongside the application proofs.

Iris-Lean [7] provides another foundation: the Iris base logic, a language-parametric program logic, and an interactive proof mode. The paper demonstrates verification and adequacy for HeapLang. Using this framework for our C library would require an instance for the C/Clight semantics, verified rules for its memory operations and calls, and an application of the adequacy theorem to those executions. Our development does not use Iris-Lean; we have not supplied that integration.

CLean’s existing contracts and function-pointer rules already support some modular reasoning. Its ownership model uses a single permission per byte, and its closure argument uses termination measures. VST supplies a broader framework with step-indexed assertions, VSU composition, and extensive proof automation. A C program can implement its own object system with function pointers and hidden mutable state. Our direct quadrature proofs do not establish the specification and automation infrastructure needed for such programs or for concurrency. Extending an existing Lean program logic to those uses is a separate project.

5.1 Calls to the original C functions

We parse the original C text and select five functions: `gauss_point`, `gauss_weight`, `integrate`, `testfun`, and `integrate_testfun`. Lean checks equalities that identify their parsed and typed syntax and the 110 table initializers with the independently imported Clight records. The parser

handles this selected part of the source and does not interpret the includes or the unrelated two-dimensional functions.

The generic call proofs separate what the integrator establishes from what its caller must supply. Each complete call includes allocation, parameter binding, expression evaluation, table loads, the loop, and return with frame release. The callback proof follows its C body, so the external-cosine variant retains a contract for cosine.

For the two-point wrapper, write $P(m)$ for the required table cells, function lookups, and cosine-call contracts in the initial memory m . The theorem has the shape

$$P(m) \implies \exists m', v. \text{Call}(\text{integrate_testfun}, m, m', v) \wedge \text{Finite}(v) \wedge |\text{real}(v) - \sin 1| \leq 0.00356. \quad (9)$$

The conclusion also identifies v with the floating-point fold and states which memory properties the call preserves. Global initialization and the chosen cosine implementation must establish $P(m)$. The polynomial library below supplies those proofs. Equation (9) describes the existence of a terminating strategy execution. Our total-correctness proof establishes the stronger property that every permitted small-step execution terminates with the specified result.

5.2 Replacing the external cosine call

The replacement starts from the degree-14 Taylor polynomial

$$P_{\cos}(x) = \sum_{k=0}^7 \frac{(-1)^k x^{2k}}{(2k)!}.$$

The C function stores binary64 approximations to its eight coefficients. It first assigns the rounded square of the argument to $\mathbf{x}$, then evaluates the polynomial in that value by Horner’s rule. Lean checks the decimal literals and the complete parsed body. The execution proof covers allocation, parameter binding, the assignment, every rounded product and sum, the return cast, and freeing. For every binary64 input, the C function returns exactly the value of the corresponding floating-point model and preserves the caller’s memory. The numerical certificates establish the approximation accuracy needed at the inputs used by the quadrature example.

The library constructor reads the five original functions and both tables from the retained C text, and reads `cos` from the replacement file. It installs six function definitions, allocates and initializes the two tables, and proves that every function lookup and table load required by the call proofs succeeds. The wrapper theorem therefore has no table or cosine contract left as a hypothesis. Starting with these source texts, it constructs the library and an initialized call returning the finite value `0x3fead02c771c35ed`, whose distance from $\sin 1$ is at most 0.00356. The integrator theorem gives the corresponding initialized calls and error bounds for every stored order from one to ten. Both the wrapper and integrator theorems cover every permitted evaluation order.

5.3 Why the result holds in every evaluation order

In the loop body, C may evaluate the weight accessor before or after the node accessor and callback. The assignment’s location can also be computed before or after its right-hand side. A proof that chooses one order does not establish what happens in the others. Our small-step semantics retains all these choices, including an explicit transition to a stuck state when an exposed subexpression is unsafe.

We define a total-correctness judgment with two cases. A state already satisfies the postcondition, or it has a successor and every permitted successor satisfies the same judgment. Each step must be

silent for these library calls. The definition is inductive, so an infinite chain of steps cannot satisfy it. The successor requirement excludes getting stuck. At the library boundary the postcondition is a terminal return with the specified value and preserved caller memory.

The central memory fact is that nested calls leave the caller’s existing loads and permissions unchanged. They may allocate and free their own local blocks. Different operand orders can assign different block numbers to those temporary allocations, so the proof does not require identical final heaps. It requires the same returned value and the same preservation property. Table reads and callback arguments therefore retain their values while another operand calls a function. The compound assignment updates only the accumulator, and the postincrement updates only the loop index. The loop invariant then gives the same rounded left fold in every order.

Theorem 5.1 (Total correctness of the initialized C library). *Let $1 \leq n \leq 10$, and let F_n be the binary64 fold of the stored n -point rule with the verified polynomial callback. From the library’s initialized memory, every execution of the original integrator in the adapted C small-step semantics terminates, returns F_n , and preserves the caller’s memory. The result is finite and satisfies the error bound in Table 3. The original two-point wrapper returns `0x3fead02c771c35ed` with integral error at most 0.00356.*

The theorem has no table or cosine-call hypothesis. Lean checks the source selection and initialization as part of its application.

The shared total-correctness lemmas handle pure expressions, calls, assignments, and control flow; the function proofs supply the table facts and loop invariant. This gives us a reusable basis for direct proofs. CLean’s separation logic offers an existing route to specifications and frame reasoning for further functions. An Iris-Lean instance for Clight would offer a different foundation. Either route would need application rules and automation suited to the programs being verified.

5.4 Preservation under C-to-Clight normalization

The frontend moves C parameters and locals into Clight temporaries and separates nested calls into statements. We prove that this translation preserves the selected library calls. Both outputs come from the same two source files, including the replacement cosine. In particular, the translated cosine is the actual frontend output: it keeps the source’s assignment to `x`, signed coefficients, and rounded Horner expression.

The proof uses the common functional result. The C total-correctness theorem covers every operand order; a separate Clight execution proof reaches the same value. All functions in the normalized library are internal. An invariant keeps every reached state in this internal fragment, where transitions are silent and deterministic under any interpretation of external calls. The finite Clight execution therefore determines every execution, without an external-determinism assumption.

Theorem 5.2 (Preservation of the selected C calls). *Let C and L be the typed C and normalized Clight libraries obtained from the retained quadrature source and the shipped cosine. Both initializers succeed with the same memory. For every stored accessor cell, every binary64 argument to the callback or cosine, every integrator call with the initialized callback and $0 \leq n \leq 10$, and the original two-point wrapper,*

$$\text{Returns}_C(\text{name}, \text{args}, t, v) \iff \text{Returns}_L(\text{name}, \text{args}, t, v).$$

Here a return includes initialization, lookup of the named function, and execution to a return state with trace t and value v . Both calls terminate without getting stuck. Their returned memories agree on every load and permission on blocks allocated before the call.

For the wrapper, both sets of returns contain exactly the empty trace and the binary64 value

0x3fead02c771c35ed. Equality of their entire final memories would be too strong: C allocates and frees parameter and local blocks, while Clight uses temporaries. The memory relation preserves what the caller could already access.

The theorem covers the selected calls in the two translated Lean representations. A general theorem about the source translator would have to justify parsing and elaboration against an independent definition of C source meaning, including preprocessing and local static arrays. That theorem remains open, as does agreement between these Lean execution rules and their Rocq counterparts.

6 Checking compiled applications

CompCert can compile a supported C program in place of GCC, and its theorem says how the behavior of the compiled program relates to that of the source. To use it for quadrature accuracy there must first be a theorem saying that the source computation is accurate, and that theorem must describe C in the semantics consumed by CompCert’s theorem. The verified core begins with formal C syntax after parsing and elaboration and ends at formal assembly syntax [16]. Schematically,

$$\underbrace{C \models S}_{\text{source correctness}} \quad \underbrace{\text{compile}(C) = A \implies \text{Beh}(A) \preceq \text{Beh}(C)}_{\text{compiler preservation}}.$$

Here S is the numerical specification and $\preceq$ is the behavior relation used by the compiler theorem. Our Lean numerical theorem concerns FloatLib’s function F , while CompCert’s theorem concerns a C program C in Rocq. A source proof must identify the computation of C with F . Reusing that proof across Lean and Rocq also requires agreement about the values, states, and propositions it describes. Using CompCert discharges the compilation step once source correctness is available in the matching semantics.

We pursue two approaches: direct proofs of the compiled applications in Lean, and application proofs in Rocq that use CompCert’s general compiler theorem. Figure 1 shows where they connect to the numerical proofs and where a connection is still missing.

6.1 Proving the compiled programs directly in Lean

For the direct Lean proofs, we ran the ten polynomial applications through CompCert and exported selected intermediate programs: Cminor, CminorSel, the initial RTL and the RTL immediately before register allocation, LTL, Linear, Mach, and formal x86-64 assembly. Each exported program is a Lean term. For each intermediate language we wrote the transition rules and an executable checker over the same FloatLib-adapted value and memory models used for Clight. We then proved, for each of the ten programs at each stage, that execution from the initial state terminates with the certified result annotation and exit status zero. Every reachable state can finish, and no execution runs forever or gets stuck.

At formal assembly an order- n run takes $153n + 37$ semantic steps. The Mach proofs check the generated return addresses against the translated continuations at all sixteen call sites. We also relate selected parts of adjacent stages, such as the contents and permissions of the global memory blocks in Clight and Cminor.

The result is a proof of each imported program’s behavior in the Lean semantics. A general compiler theorem would have to relate the input and output of every successful compilation. To reuse CompCert’s theorem in Lean, we would need to relate the exported syntax and adapted semantics to their Rocq definitions, then justify interpreting the Rocq theorem in Lean. Another

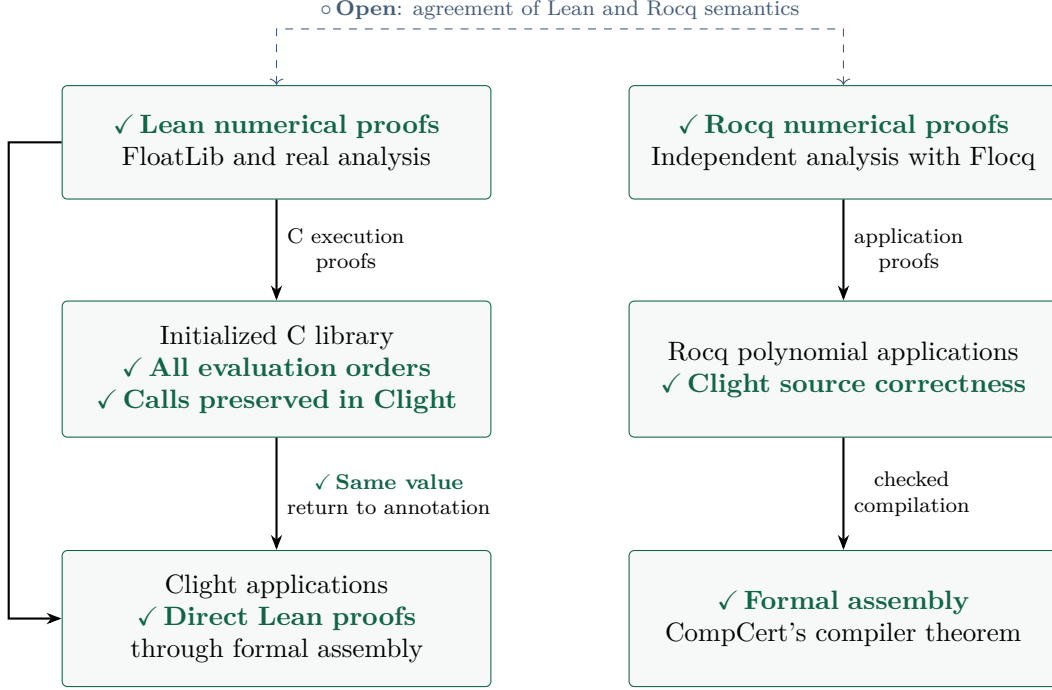


Figure 1: The two developments within LeanQuadrature. Solid arrows connect proved results; dashed arrows mark missing connections. Lean numerical proofs support the initialized C library and applications checked through assembly. The left vertical arrow compares the value returned by C with the value reported by the application; it does not assert a general compilation theorem. The right column describes our independent Rocq proofs. Both columns end at formal assembly syntax, before native assembly and linking.

approach would be a translation validator that checks each compilation and has a proved soundness theorem.

6.2 From a C return to an application result

The library returns a `binary64` value, while the application reports one and exits. To relate them, let $R_n(v)$ mean that the initialized C integrator, called with the polynomial callback and order n , returns v with an empty trace. Let $A_n(t, s)$ mean that the corresponding initialized assembly application terminates with trace t and exit status s , and write $T(v)$ for the single `quadrature-result` annotation carrying v .

Theorem 6.1 (Agreement of C returns and assembly observations). *For every $1 \leq n \leq 10$, trace t , and exit status s ,*

$$A_n(t, s) \iff \exists v, R_n(v) \wedge t = T(v) \wedge s = 0.$$

Both the C call and the assembly application terminate without getting stuck. For the original two-point wrapper, the returned and reported value has bits `0x3fead02c771c35ed` and integral error at most 0.00356.

The proof compares each side with the same FloatLib result F_n . The initialized C call can return only F_n , and the assembly application's only final observation is $T(F_n)$ with status zero. The existing execution proofs establish that both outcomes occur and that every execution terminates.

These statements hold for any external-call relation: the library calls are internal, and the assembly execution uses only the specified annotation.

A companion theorem relates returns from the actual normalized Clight library to the authored Clight application’s observations. Their cosine bodies differ syntactically but evaluate the same rounded polynomial. That theorem retains the Clight application proof’s external-call determinism premise. The direct C-to-assembly result above does not need that premise.

This agreement supplies the result connection for these applications. Their entry points were written in Clight, and the imported assembly trees were checked directly. A proof that a compiler applied to the original C text produces these trees would be a separate result, as would a theorem relating the adapted Lean semantics to Rocq’s.

6.3 Using CompCert in Rocq

The second route uses CompCert’s theorem in Rocq. We reproved the numerical results for the polynomial programs in Rocq, independently of the Lean proofs, and composed them with compilation and termination proofs. All ten orders have integral-error certificates whose conclusions hold for every behavior of the compiled program. Successful compilation is itself checked inside Rocq, so it does not remain a hypothesis.

Evaluating CompCert inside the Rocq kernel needs a configuration that the kernel can run. We supply executable definitions for the compiler’s external choices, disable selected optimizations and inlining, and use a finite set of register-allocation candidates that CompCert’s validator must accept. The configuration also exposes computations needed by evaluation and uses a bounded iteration procedure whose invariant theorem we reprove. The certificate script rebuilds the compiler proofs from source. Its finite allocator may fail on other programs, so the results describe this checked compiler variant and these applications.

6.4 Connecting the two proof systems

The independent Rocq proofs supply the numerical premises for the CompCert applications. Reusing the Lean proof would avoid repeating that analysis. One route is a sound interpretation or translation of the relevant arithmetic, program states, and propositions. Another is a common certificate format with sound checkers in both systems. Either approach must justify the interpretation of the checked result in the receiving system.

Floating-point arithmetic shows what has to be related. The FloatLib and Flocq models used here have different NaN policies, and paired certificates show that unrestricted bitwise equivalence between them is false. For quadrature the useful comparison concerns finite computations and must preserve the sign of zero. We have proofs in both systems that characterize rounding through integer quotients, remainders, parity, and exponent alignment. Relating those integer representations is still open.

6.5 From formal assembly to an executable

The CompCert manual places assembly printing, assembling, and linking outside the verified core. Our final property concerns a `quadrature-result` semantic annotation and exit status zero in formal assembly. The printer turns the annotation into a comment, so this property does not yet describe a printed answer from a native executable. Connecting it to one requires proofs about the intended output operation and the remaining translation steps. CompCert could also compile C emitted by Lean’s own compiler. Its theorem would then cover the C compilation step, and the Lean-to-C translation and runtime would still need justification.

7 Results

Table 3 gives the results for the ten polynomial applications. Each row identifies the binary64 answer and a proved bound on its distance from $\sin 1$. Lean and our Rocq development check these bounds independently. At the higher orders the bounds are on the scale of rounding and stored-table perturbations, and increasing the number of nodes does not make these finite-precision bounds decrease at every step: the seven-node bound is smaller than the eight-node bound.

Nodes	Binary64 result bits	Error at most	Status
1	3ff0000000000000	0.159	✓ Proved
2	3fead02c771c35ed	0.00356	✓ Proved
3	3feaed9520c8a014	$3.1 \cdot 10^{-5}$	✓ Proved
4	3feaed5443a06327	$1.5 \cdot 10^{-7}$	✓ Proved
5	3feaed548f3f6f46	$4 \cdot 10^{-10}$	✓ Proved
6	3feaed548f08f234	$8 \cdot 10^{-13}$	✓ Proved
7	3feaed548f090ce1	$2 \cdot 10^{-15}$	✓ Proved
8	3feaed548f090cce	$4 \cdot 10^{-15}$	✓ Proved
9	3feaed548f090ccf	$4 \cdot 10^{-15}$	✓ Proved
10	3feaed548f090cd4	$3 \cdot 10^{-15}$	✓ Proved

Table 3: Proved error bounds for the ten initialized polynomial applications. The callback evaluates the degree-14 cosine polynomial with fixed rounding at each operation.

LeanQuadrature [9] contains the application proofs, the build instructions, and a map from every result in this paper to its theorem declaration. The full Lean build and transitive assumption checks cover the project theorems, their private helpers, and the adapted CLean library, and the only permitted axioms are `propext`, `Classical.choice`, and `Quot.sound`. The Rocq compilation proofs have their own record of assumptions, including the logical and external-semantics dependencies of CompCert and Flocq. The project contains roughly 50,000 lines of Lean, a large share of it imported program syntax, and about 10,000 lines of Rocq. LeanPDE is a private dependency. The public source distribution does not include it; readers who wish to reproduce the build should request access from the author.

8 Discussion

The mathematical and numerical proofs draw heavily on existing Lean libraries. `mathlib` provides the analysis needed for the general Gaussian construction; `LeanPDE` contributes quadrature interfaces, concrete rules, and certified symbolic calculation; `FloatLib` makes the binary64 computation executable inside the logic. `LeanPDE` contains the weighted construction, remainder, convergence, and Legendre certificate proofs. `LeanQuadrature` applies these results to certify the ten stored rules and prove the programs that use them. `LeanPDE` is essential to this implementation because we import those definitions and theorems. `TorchLean` is a transitive dependency of `LeanPDE`; we do not use its tensor or neural-network interfaces.

The larger difference between the two systems appears in program verification. Rocq supplies VST’s C program logic and CompCert’s compiler theorem in the same proof system. Lean already has CLean’s Clight contracts, separation logic, function-pointer rules, and module-linking operations. Our application proofs use its operational semantics directly. Iris-Lean offers a further program-logic

foundation, whose use here would require a C/Clight language instance and verified rules for memory operations and calls. The existing Lean infrastructure is substantial; the present development does not establish VST’s broader modular reasoning and automation, or a transfer of CompCert’s compiler theorem.

The source review gives concrete corrections to Appel and Bindel’s preliminary application argument. The two error bounds and three admitted helpers have counterexamples; the remainder needs consistent indexing and a smoothness premise; the table model must agree with the initialized C values. The callback body proof and the incorrectly named Hughes specification are separate program obligations. Our Lean proofs supply corrected mathematical statements and verify the polynomial replacement. The public derivative-bound work already moves toward stronger hypotheses, and the missing quadrature VSU explains why the initializer and function-identifier mismatches are not caught by the existing body proofs. These are unfinished steps in the authors’ development, not failures of VST’s program logic. The overflow inequalities were transcribed into Lean and checked against the source by inspection. The positivity counterexample was also checked against the original statement in Rocq.

The C library’s return values are now related to the application observations through formal assembly. This is a result about the particular verified programs, with an explicit change from a library return to a result annotation. A general source-translation theorem would still have to justify parsing and elaboration, including preprocessing and static-array handling, and a compilation theorem would have to relate compiler inputs to outputs. Reusing CompCert’s compiler theorem in Lean adds a different obligation: proving a sound interpretation of the Rocq semantics and results. A theorem about a native executable’s printed answer must also account for assembly printing, assembling, linking, and the output operation.

A Gaussian quadrature in LeanPDE

LeanPDE supplies the Gaussian theory used here: weighted rules of every positive order, orthogonal polynomials, roots, exactness, positive weights, recurrence, remainder, and convergence. It also provides explicit Legendre rules through four nodes and general root and weight certificates.

Table 4 follows the 24 numbered paragraphs of Stewart’s *Afternotes on Numerical Analysis*, Chapter 23, as used in Appel and Bindel’s Section 2 [15]. The left column states the mathematical step; the right names the LeanPDE result and explains its proof or scope. Their PDF skips paragraph 14, whose root argument appears in the `quadrature.v` comments.

For the compact-interval results, assume finite $a < b$, a continuous weight $w > 0$ on $[a, b]$, and $n > 0$ nodes. Write $\mathcal{I}_w$ for the weighted integral, p_n for the degree- n monic orthogonal polynomial, and ℓ_i for a cardinal Lagrange polynomial. Stewart’s indices $0, \dots, m$ correspond to $n = m + 1$, so his derivative order $2m + 2$ becomes $2n$ here.

LeanPDE also gives the sharp two-point estimate

$$\left| \int_a^b f(x) dx - Q_2(f) \right| \leq \frac{M(b-a)^5}{4320}, \quad |f^{(4)}(x)| \leq M,$$

under $a < b$ and $f \in C^4([a, b])$. On $[-1, 1]$ this is $M/135$. The theorem is `norm_integral_sub_gaussLegendreTwoRule_valueOn_1e` in LeanPDE’s `GaussLegendrePeano` module. The general Hermite remainder supplies the same coefficient and also gives a single point witnessing the integrated remainder identity. The companion [LeanPDE source map](#) gives file links and precise hypotheses for the results below.

Table 4: Stewart’s Chapter 23, as followed in Appel and Bindel’s Section 2: the 24 mathematical steps and their current LeanPDE coverage. The statements use n nodes. Checkmarks mark available definitions and proved results; alternative arguments and open cases are identified.

¶	Textbook statement or argument	LeanPDE result and proof
1	Approximate the weighted integral $\mathcal{I}_w(f)$ by a sum of n values, $\mathcal{Q}_n(f) = \sum_i \lambda_i f(x_i)$.	✓ Defined: <code>GaussianRule</code> and <code>GaussianRule.value</code> represent and evaluate the rule. Its existence is proved using the orthogonal polynomials and their roots below.
2	Keep the interval and weight fixed, and abbreviate integration against w by $\mathcal{I}_w$.	✓ Defined: <code>polynomialIntegral</code> is the weighted integral restricted to polynomials.
3	Integration commutes with addition and scalar multiplication for integrable functions.	✓ Defined and proved: <code>polynomialIntegral</code> is a linear map, using <code>mathlib</code> ’s integral linearity.
4	Define orthogonality by $\mathcal{I}_w(pq) = 0$, using the integral of the product as an inner product.	✓ Proved: <code>momentInnerCore</code> constructs the polynomial inner product; <code>polynomialIntegral_square_pos</code> proves positivity on nonzero squares.
5	Choose a monic polynomial p_k of each degree k , with $\mathcal{I}_w(p_i p_j) = 0$ whenever $i \neq j$.	✓ Proved: <code>exists_monic_orthogonal</code> constructs the family by orthogonal projection; <code>orthogonalPolynomial_unique</code> proves uniqueness at each degree.
6	Every polynomial of degree at most n has a unique expansion in any monic family $p_0, \dots, p_n$ with $\deg p_k = k$.	The arbitrary-family expansion theorem is not separately formalized. Orthogonal projection gives the family directly, so the Gaussian construction does not need this intermediate result.
7	Prove that expansion by subtracting the leading multiple of p_n , lowering the degree, and applying induction.	<code>degree_sub_coeff_mul_lt</code> proves the degree reduction used in the recurrence. It does not establish the full expansion and uniqueness argument of paragraph 6.
8	An orthogonal polynomial p_n is orthogonal to every polynomial of degree below n .	✓ Proved: <code>orthogonalPolynomial_orthogonal</code> . This is a defining property of the projection construction.
9	The family starts with $p_0 = 1$ and $p_1(x) = x - \mathcal{I}_w(x)/\mathcal{I}_w(1)$, as determined by orthogonality to constants.	✓ Proved: <code>orthogonalPolynomial_zero</code> and <code>orthogonalPolynomial_one</code> , including the weighted mean in the degree-one coefficient.
10	Inner products determine the coefficients of p_n and p_{n-1} when expressing $x p_n - p_{n+1}$ in the orthogonal family.	✓ Proved: <code>recurrenceAlpha</code> and <code>recurrenceBeta</code> give those ratios of integrals; the recurrence theorem proves the identity.
11	All earlier terms vanish: $x p_j$ has degree below n when $j < n - 1$, so it is orthogonal to p_n .	✓ Proved: lower-degree orthogonality removes these terms in the proof of the three-term identity. No separate infinite coefficient list is needed.

Table 4 continued: Stewart’s Chapter 23 and Appel–Bindel Section 2.

¶	Textbook statement or argument	LeanPDE result and proof
12	After the first two polynomials, each p_{n+1} is obtained from $x p_n$, p_n , and p_{n-1} by a three-term recurrence.	✓ Proved: <code>orthogonalPolynomial_recurrence</code> , with <code>recurrenceBeta_eq_norm_ratio</code> and <code>recurrenceBeta_pos</code> . The recurrence is derived after the projection construction.
13	The degree- n orthogonal polynomial has exactly n real, distinct roots, all inside (a, b) .	✓ Proved: <code>orthogonal_splits</code> , <code>orthogonal_squarefree</code> , and <code>orthogonal_root_mem_Ioo</code> give splitting, simplicity, and interiority.
14	Multiply p_n by the product of the linear factors at its sign-changing roots. Too few such roots would make orthogonality contradict a nonzero integral.	✓ Alternative proof: <code>orthogonal_factor_has_root</code> forces each nonconstant factor to have an interval root. Factorization gives splitting; separate positivity arguments give simplicity and exclude endpoints.
15	Use the roots as nodes and set $\lambda_i = \mathcal{I}_w(\ell_i)$. Lagrange interpolation then gives exactness below degree n .	✓ Proved: <code>interpolatoryWeight</code> and <code>interpolatory_exact</code> ; <code>GaussianRule.weights_eq_interpolatoryWeight</code> identifies the Gaussian weights with these integrals.
16	For $\deg f \leq 2n - 1$, divide $f = p_n q + r$ with $\deg q, \deg r < n$. Orthogonality and interpolation give $\mathcal{Q}_n(f) = \mathcal{I}_w(f)$.	✓ Proved: <code>gaussian_exact</code> follows this division argument. The $p_n q$ term vanishes at the nodes and has zero weighted integral; the rule integrates r exactly.
17	Apply exactness to ℓ_i^2 . Its quadrature value is λ_i , and its weighted integral is positive.	✓ Proved: <code>gaussian_weight_pos</code> uses this squared-Lagrange argument to prove every weight positive.
18	The weights sum to $\mathcal{I}_w(1)$. The quoted prose also concludes that each weight is at most one.	✓ Proved: <code>sum_weights</code> and <code>weight_le_mass</code> . The general bound is $\lambda_i \leq \mathcal{I}_w(1)$. The Rocq theorem already states this mass bound; the bound one follows under unit-mass normalization.
19	For sufficiently smooth f , the quadrature error is $\mathcal{I}_w(p_n^2)$ times $f^{(2n)}(\xi)/(2n)!$ at some interval point ξ .	✓ Proved: <code>hermite_remainder</code> , <code>GaussianRule_remainder</code> , and <code>GaussianRule.abs_error_le</code> . Hermite interpolation gives the pointwise error; the integrated identity uses the continuity assumptions in Section 3.
20	Approximate a continuous function uniformly by polynomials. Positivity and increasing exactness then give $\mathcal{Q}_n(f) \rightarrow \mathcal{I}_w(f)$.	✓ Proved: <code>gaussian_rules_converge</code> follows this uniform-approximation argument on a compact interval with fixed positive continuous weight.
21	Choose $w = 1$ on $[-1, 1]$. The resulting family consists of the monic Legendre polynomials and yields Gauss–Legendre quadrature.	✓ Proved: the general construction specializes to unit weight. Explicit rules through four nodes are identified with the monic Legendre family. General root and weight certificates support the stored tables.
22	Gauss–Laguerre quadrature treats an integral on $[0, \infty)$.	◦ Open: the corresponding unbounded-interval construction, remainder, and convergence theory are outside this development.

Table 4 continued: Stewart’s Chapter 23 and Appel–Bindel Section 2.

	Textbook statement or argument	LeanPDE result and proof
23	Gauss–Hermite quadrature treats an integral on the whole real line.	◦ Open : this unbounded-interval quadrature theory is outside the development. Hermite interpolation in paragraph 19 is a different result.
24	Other choices of interval and weight produce Gaussian rules suited to other integrals.	exists_gaussian_quadrature applies to any positive continuous weight on a compact interval. Individual named families require their own specialization and certificates.

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